

Fossils and Capitalism: Laws and Ethics of the Commercialization of Paleontological Resources

English 2010 Persuasive Research Essay

Caleb Young – A02362954

On December 9, 2022 (Jacobs), a large *Tyrannosaurus rex* Skull – named Maximus (fig. 1) – was sold on Sotheby's (an auction house in New York) to an anonymous private buyer for \$6,069,500 ("Tyrannosaurus rex Skull: Maximus rex"), much to the dismay of many paleontology fans. The fossil may have sold for well under the estimated \$15-20 million, however much of the controversy comes from the fact it was sold at auction in the first place. Many wish it could have



Figure 1. Picture of the fossil for sale. *Tyrannosaurus rex* Skull, Sotheby's, www.sothebys.com/en/buy/auction/2022/maximus-rex/tyrannosaurus-rex-skull. Accessed 13 Dec. 2022.

been donated to a museum so it could more easily be studied by paleontologists and viewed by the public. Joe Dziemianowicz, a journalist at Barron's Penta, wrote, quoting The University of Maryland's Vertebrate Paleontologist, Thomas R. Holtz, "One of the main issues with the sale of dinosaur or other fossils as trophies and collectible objects is that it misses out on a key aspect of the specimens—and that's data." As the fossil sits on display in a private collection, access to it will be limited. In many other countries, the sale of a fossil to an individual or potentially moving it outside the country is prohibited, but such restrictions don't apply in the United States

if the fossil is found on private land. Should the U.S. government enact laws to discourage the selling of these invaluable paleontological resources, such as the fossil Maximus?

To better understand what, if anything, should be done, a better understanding of what has been done (or what current laws are in place) will help provide some framework. The collecting of fossils on federal land is restricted under the Paleontological Resources Preservation Act. The act states, “a paleontological resource may not be collected from Federal



Figure 2. A fossilized shark tooth (likely Great White). Young, Caleb A. *Fossil Shark Tooth*.

land without a permit issued under this subtitle by the [Secretary of the Interior],” the only exception being “casual collecting,” defined as, “the collecting of a reasonable amount of common invertebrate [such as ammonite] and plant [such as petrified wood]

paleontological resources for non-commercial personal use.” While not specifically stated, common parts of vertebrate animals, such as shark teeth (fig. 2), are generally considered to fall under casual collecting as well. State laws may further restrict collecting on state lands, but neither cover fossils found on private lands such as Maximus; Sotheby’s states the fossil was found in the Hell Creek formation in South Dakota on privately owned land. These fossils are owned by whoever owns the land, and that person can do with the fossils what they want, whether it be destroy them or allow people to excavate them. Scientific institutions have difficult competition when it comes to excavation: commercial fossil hunters. “Bidding wars can drive up a fossil’s price to far beyond what any university or museum could afford,” writes National Geographic journalist Michael Greshko in his article, “Stan the T. rex

just sold for \$3.8 million-and scientists are furious”, “especially. . . as COVID-19 has financially devastated museums the world over.”

The sale of fossils in auction houses suggests they are viewed more as works of art than something of scientific importance (Gramling). While, like a work of art, each specimen is unique, that’s where similarities end. As Sumida puts it, “All fossils are data.” While certainly not every one of the millions of Ammonite fossils is worth studying, for rarer species like *Tyrannosaurus*, each one is of great significance. The problem with the commercialization of fossils lies not in who exactly owns what fossil, but the limited access that comes with private ownership. Fossils owned in private collections can’t be viewed by the public, and often not by scientists either. Even when they can be studied, valuable information can often be lost before paleontologists can get to it, as fossil hunters are typically only concerned with the specimen itself, and not its surroundings (Roddy 479). Maximus isn’t the first *Tyrannosaurus* to be sold at auction to a private collector. The few specimens of the species available for study does not make those hidden away in private collections any less of a blow, either. According to Carolyn Gramling, Earth and Climate Writer for Science News, “There are only about 120 known specimens of *T. rex* in the world. At least half of them are owned privately and aren’t available to the public.” Data from the other half of the specimens would allow scientists to better understand much about *Tyrannosaurus*, including how it grew or differences based on its sex (Gramling). Gramling states, paraphrasing California State University’s Paleontologist Stuart Sumida, “[he] is particularly appalled at statements made by auction houses that suggest the skeletons “have already been studied,” an attempt to reassure researchers that the data contained in that fossil won’t be lost, regardless of who purchases it. That’s deeply misleading, he says, because of the need for reproducibility, as well as the always-improving development of new analysis

techniques. “When they make public statements like that, they are undermining not only paleontology, but the scientific process as well.””

Allowing constant access to the fossils allows new findings to be compared to old ones, and for studies to be reproduced to determine their validity. In 1912, Freiherr Ernst Stromer von Reichenbach and Richard Markgraf discovered portions of a new dinosaur they named *Spinosaurus aegypticus*, which was soon housed in Germany’s Bavarian Museum (“Rediscovering Spinosaurus, the Lost Dinosaur”). With the start of World War II, Stromer

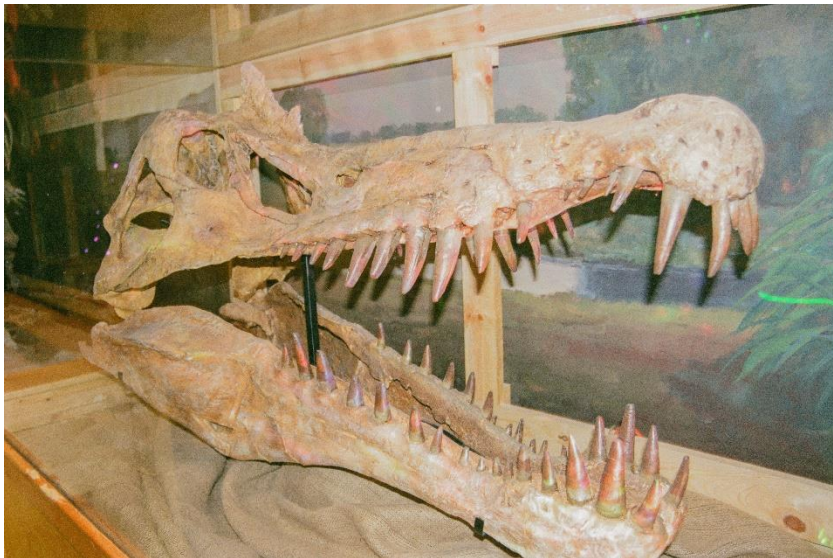


Figure 3. Spinosaurus Skull on Display at the Ogden George S. Eccles Dinosaur Park. Young, Caleb A. Spinosaurus Skull.

became concerned that the fossils could become damaged, and pleaded with the head of the museum “to move the collection from Munich to the safety of caves and salt mines” (“Rediscovering Spinosaurus, the Lost Dinosaur”). Despite

his pleas, however, the fossils were left in the museum where they were destroyed by a bombing in 1944, leaving only a few sketches, notes, and photographs (“Rediscovering Spinosaurus, the Lost Dinosaur”). While we are lucky that these sketches remain, the destruction of Spinosaurus (and countless other specimens in the museum) is still tragic. New methods of investigation (such as determining if soft tissue could still be found inside), as well as comparisons with newer findings of Spinosaurus, are limited or nearly impossible. Fossils sitting in private collections may as well have been bombed with how useful they are to science. Rather than being available

for further studies and comparison with new findings, they sit as conversation pieces in the mansions of the rich.

While it may seem like banning private collecting entirely may help, Bridget Roddy, author of “Can You Dig it? Yes, You Can! But at What Cost?” disagrees, stating, “The commercial fossil business has allowed for more paleontological discoveries to reach the public and is a major and necessary source of fossils for museums. . . If new legislation were to eliminate the industry entirely, then fewer fossils would be available for everyone.” She also argues that the government seizing all fossils under eminent domain is equally unhelpful, “Even if legislators could pass a vesting statute, this would require the government to pay “just compensation” under eminent domain for all material seized from private land. Because of the vast number of fossils in the United States, a statute of this magnitude would require a significant amount of federal funds. This would be an ongoing and unusually unjustifiable expense because a substantial amount of material unearthed is not of scientific interest.” If these aren’t viable solutions, then what is?

One potential solution to curb the commercialization of fossils is extending the same protection over them that other archeological artifacts have. Roddy uses the Aboriginal Mounds, Earthworks and Other Antiquities Law in Alabama as an example, stating, “This law reserves the exclusive right and privilege of the state to “explore, excavate or survey aboriginal mounds, earthworks, burial sites, etc.” and “state ownership of objects found or located therein declared.” This act goes beyond reserving these rights solely on state-owned land by including privately owned land, “subject to the rights of the owner of the land upon which such antiquities are situated.”” If extended to cover fossils as well, the state could seize ownership of significant fossils, leaving those of little scientific worth to whomever discovered them (Roddy 504). While

helpful, each individual state would have to enact similar laws for it to be fully effective, or at the very least those states which are most rich in fossils, such as Utah, Montana, and California.

Another potential solution Roddy mentions is a tax on paleontological resources, similar to the tax many states have on items like Cigarettes (498). A large tax on these resources could be just what is needed to give museums and other scientific institutions the upper hand on these auctions. “This type of sales tax would give museums a competitive advantage in public sales of scientifically important finds by waiving a sales tax and by incentivizing private buyers to not drive-up auction prices by imposing an additional tax to the auction price,” she says (498). While it may not prevent someone from keeping fossils they find themselves, it would help place more specimens in the hands of the scientific community and in the eyes of the public. Furthermore, the additional revenue from the taxes could be used not just to enforce the laws, but also to fund more research on these resources (Roddy 497). While laws like these may be helpful, they don’t fully protect fossils from exportation.

Many countries with important paleontological resources, have laws prohibiting the exportation of fossils. With the US being so rich in fossil specimens, it’s odd that the country doesn’t follow suit. Roddy proposes that the country treat fossils similarly to how Italy treats its antiquities, passing laws that prevent or restrict the movement of fossils to outside the country (Roddy 489). Keeping the fossils in the country where they were found will help preserve the history of the land for generations to come. While the exportation of fossils from the U.S. to other countries is legal now, the removal of fossils from Indian lands is not, yet the problem persists.

The most famous case of controversy over fossil ownership belongs to Sue, the largest and most complete Tyrannosaurus skeleton ever discovered (“Sue the T. rex.”). While the Black

Hills Institute of Geological Research was given permission to dig by the supposed owner of the land, Maurice Williams, it turns out he was a member of the Cheyenne River Sioux Tribe, and the land was tribal land owned in trust by the U.S. government (Jones 7-8). Eventually, after several legal battles, it was determined that the fossil was U.S. property, but Williams was allowed to sell it (Jones 8). Sue went to auction in 1997 where, (with the help from several generous donors) it was bought by the Chicago Field Museum for \$8.1 million (“Sue the T. rex.”), about \$7 million of which went to Williams (Jones 8). While the fossil did end up in the hands of the scientific community in the end, the high price likely drew much attention to fossils as high-ticket items and may have contributed to Maximus’ high sale price (Roddy 490).

Sue isn’t the only fossil that’s been involved in legal battles with Indian tribes in the United States; disputes go back as far as 1980 between hunters and various tribes (Dalton 858). Many tribes are hoping to extend the Native American Graves Protection and Repatriation Act (or at least its concepts) to fossils as well, requiring that they be deemed the property of the tribe they’re associated with. Throughout history, fossils have been important in the cultures of people everywhere, not least the Pawnee tribe in North America. Roddy Explains that Pawnee medicine men “undertook vision quests [to nahurac, or spirit animal, mounds] where they were said to encounter mysterious creatures and receive special healing powers (478). One such mound was Pahowa on the Solomon River in Kansas, a unique limestone formation about 40 feet high with a spring-fed mineral pool on top, described by Pawnee elders as a high timbered bank where immense, petrified bones spilled out.” These tribes may still hold fossils in high regard to this day. Unlike those auctioned off to private collections, it seems the fossils the Indian tribes are claiming will be more accessible, at least if the Dakota/Lakota people of Standing Rock are anything to go by. They have collaborated with the South Dakota School of Mines and

Technology to develop a paleontology program, in which tribe members learn and work alongside outside paleontology experts (Dalton 859). Rex Dalton, author of “School of Rock,” states, “The 6,000-member [Standing Rock] tribe is also building a fossil-preparation laboratory at its Sitting Bull College and hopes to build a palaeontological museum to bring not only tourists and economic development, but also scientists to study specimens found on the reservation.” The program seems to be ideal for all parties involved. The tribe can profit not just financially from tourists brought to the museum, but also intellectually as Indian students learn invaluable information about paleontology. One student, a Seminole Tribe member Matthew Wood, noted, “I learned much more than I expected. It was all hands-on — an incredible experience.” With the fossils proudly displayed and available for further studies, scientists and the public who aren’t members of the tribe can still enjoy and learn from them, too.

A sale similar to that of Maximus occurred just two years prior. In October of 2020, Stan the *T. rex* sold for \$31.8 million at Christie’s auction house to a private buyer (Greshko “Scientists are Furious”). Many paleontologists were rightfully worried, among them Thomas Carr, a paleontologist at Carthage College. Greshko quoted Carr in his article “‘Stan’ The *T. rex* Just Sold for \$31.8 Million-and Scientists Are Furious” saying, “Do the right thing: totally relinquish your ownership of the fossil and donate it to an accredited natural history museum so that science can ethically be done on Stan for the benefit of everyone on the planet who has an interest in dinosaurs,” (Greshko “Scientists are Furious”). Since then, Greshko and the National Geographic team have attempted to track Stan down, publishing their findings in March 2022. In the article, titled “Stan the *T. rex* found! World’s most expensive fossil finds home in a new Museum,” Greshko explains that the anonymous buyer had Stan shipped to Abu Dhabi where it will be displayed in a new museum upon its completion, expected to be 2025 (Greshko “Stan

found”). It may seem odd that Dhabi purchased the specimen for such a high price (currently it is the most expensive fossil sale to date), but Greshko explains, “the [United Arab Emirates]’s rock deposits aren’t the right age and type to preserve the fossil world’s biggest celebrities: dinosaurs. So, the new museum had to source them from abroad.” Miraculously, Stan is still available for continuous study by paleontologists, and while Stan may have been moved from the U.S., it can still be viewed by tourists, as well as citizens of the U.A.E. who haven’t had access to specimens like these before. One can only hope that Maximus gets the same treatment, and its owner donates it to a museum where it will soon make an appearance.

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